

CS 4782 – Deep Learning

Final Project Proposal

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1 Chosen Paper

Paper URL:

<https://arxiv.org/pdf/2301.10226>

Title:

A Watermark for Large Language Models

Authors:

John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein

Affiliation:

University of Maryland

Brief Summary of the Chosen Paper

The paper proposes an algorithm for embedding a watermark in the output of large language models (LLMs) and a statistical method for detecting that watermark later. The watermark biases the model toward selecting tokens from a dynamically generated “green list” of tokens. Over long sequences, this bias produces a statistical signal that can be detected without requiring access to the original model.

Watermark Generation Process

1. The model tokenizes the input text.
2. Each token is mapped to its corresponding token ID.
3. Using a random seed derived from a secret key, the vocabulary IDs are partitioned into a *green list* and a *red list*. The green list typically contains about 25% of the vocabulary.
4. The model produces logits representing the likelihood of each token appearing next.
5. If a token belongs to the green list, a bias is added to its logit, increasing the probability that it will be selected.

6. The logits are converted into probabilities using the softmax function.
7. The next token is sampled and appended to the sequence. This process repeats until the generation is complete.

Watermark Detection Process

1. The green list fraction is determined by the parameter γ .
2. Convert the input text into tokens and map each token to its corresponding token ID.
3. Reconstruct the green list sequentially using the secret key, the previous token, and γ .
4. If the current token belongs to the reconstructed green list, increment **greenCount**.
5. Repeat this process for every token in the sequence.
6. Compute the following z -score to determine whether the text is watermarked:

$$z = \frac{G - \gamma T}{\sqrt{T\gamma(1 - \gamma)}}$$

where:

- G = number of observed green tokens (**greenCount**)
- T = total number of tokens
- γ = fraction of the vocabulary assigned to the green list

7. In the paper, a watermark is considered strongly detected when $z \geq 4$.

Justification for Choosing This Paper

The rapid rise of AI tools has increased the likelihood that students rely on large language models to complete assignments. For example, a student could submit a prompt to ChatGPT and directly submit the generated response. This reliance on AI may negatively affect the quality of learning and academic integrity. The watermarking approach proposed in this paper offers a potential solution for detecting AI-generated text and may help address this issue.

2 Data

Proof that the Data Exists / Is Accessible

The data required to replicate this experiment include:

- An open-source language model to test. We plan to use OPT-350M. The original paper used OPT-6.7B, but due to computational limitations we will use the smaller OPT-350M model.
- Model-generated text. We will generate outputs directly from the model, making this data straightforward to obtain.
- A prompt dataset. We plan to use the dataset referenced in the paper:
`load_dataset("allenai/c4", "realnewslike")`

3 Result Selection

Results We Aim to Replicate

- Watermark detection accuracy close to 95%
- False positive rate near 0%
- High true positive rate for watermarked text
- Successful generation of model outputs containing detectable watermarks

4 Re-implementation Plan

Architecture, Method, and Metrics

We will use the OPT-350M language model implemented through the HuggingFace Transformers library and run experiments using Google Colab.

Our plan is to generate approximately 2,000 sequences of 512 tokens each. Based on preliminary estimates, generating this amount of data using OPT-350M on a free Colab GPU should take approximately 3–4 hours.

The primary metrics we will measure include:

- Watermark detection accuracy
- True positive rate
- False positive rate
- Distribution of z -scores for watermarked vs. non-watermarked text

These metrics will allow us to evaluate whether the watermark detection method performs similarly to the results reported in the paper.